Mid-South Logistics

A national logistics company operates a fleet of trucks that transport goods across multiple regions. Recently, management has noticed inconsistent delivery performance, increasing customer complaints, and fluctuating inventory levels. They believe operational factors such as traffic conditions, shipment status, waiting time, asset utilization, weather conditions, and demand forecasting may be contributing to logistics delays.

The purpose of this project is to analyze operational logistics data to uncover patterns that affect delivery performance, asset utilization, and inventory management. The findings will help management make data-driven decisions that improve delivery efficiency, reduce delays, and better allocate company resources.

Problem Statement

The company is experiencing frequent logistics delays that may be increasing operational costs and reducing customer satisfaction. Management needs to understand which operational factors have the greatest impact on delivery performance so they can improve efficiency and reduce delays.

The main question I want to figure out is how can we fix the problems the company has been having to improve customer support and driver accessibility.

Inventory Analysis

	Asset_ID	Traffic_Status	Inventory_Level	Waiting_Time
1.	Truck_4	Heavy	103	22
2.	Truck_2	Detour	103	52
3.	Truck_3	Heavy	102	50
4.	Truck_10	Detour	100	22
5.	Truck_2	Detour	100	29
6.	Truck_4	Detour	100	39
7.	Truck_2	Clear	100	49
8.	Truck_6	Detour	100	43
9.	Truck_2	Detour	100	22
10.	Truck_4	Clear	100	19

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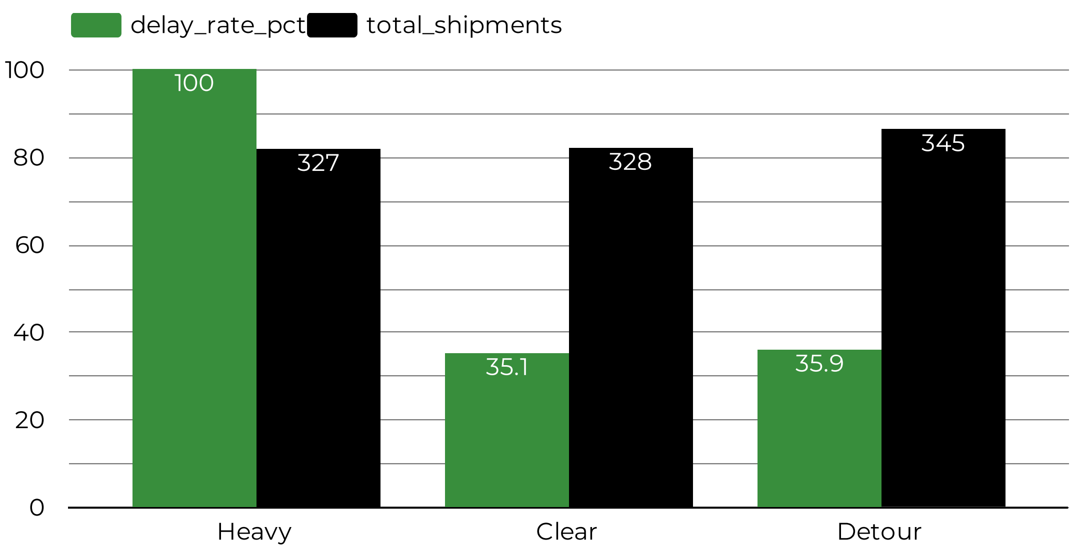
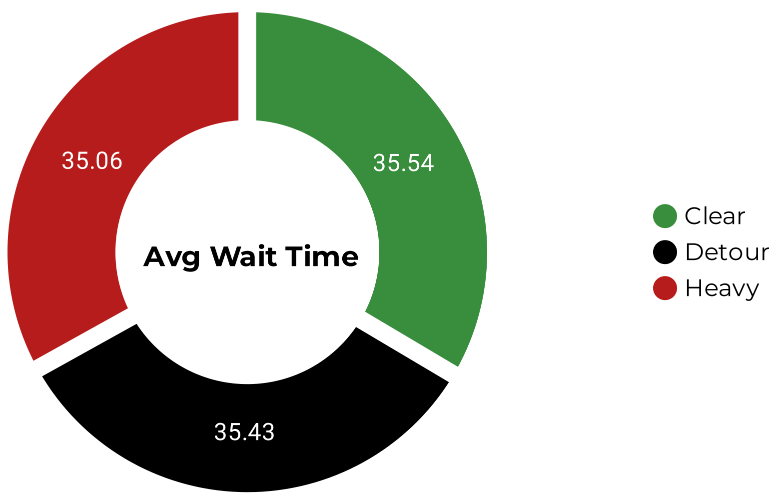
Average Inventory

297.92





From the data, we can see the average waiting time for clear traffic(35.54%) and detour traffic (35.43%) are really high. However, the heavy traffic (35.06%), still outweighs both since it still has a 100% delay rate.

Enviromental Analysis

Logistics_Delay ▼	avg_humidity	avg_temperature
1	65.04	23.78
0	65.05	24.04

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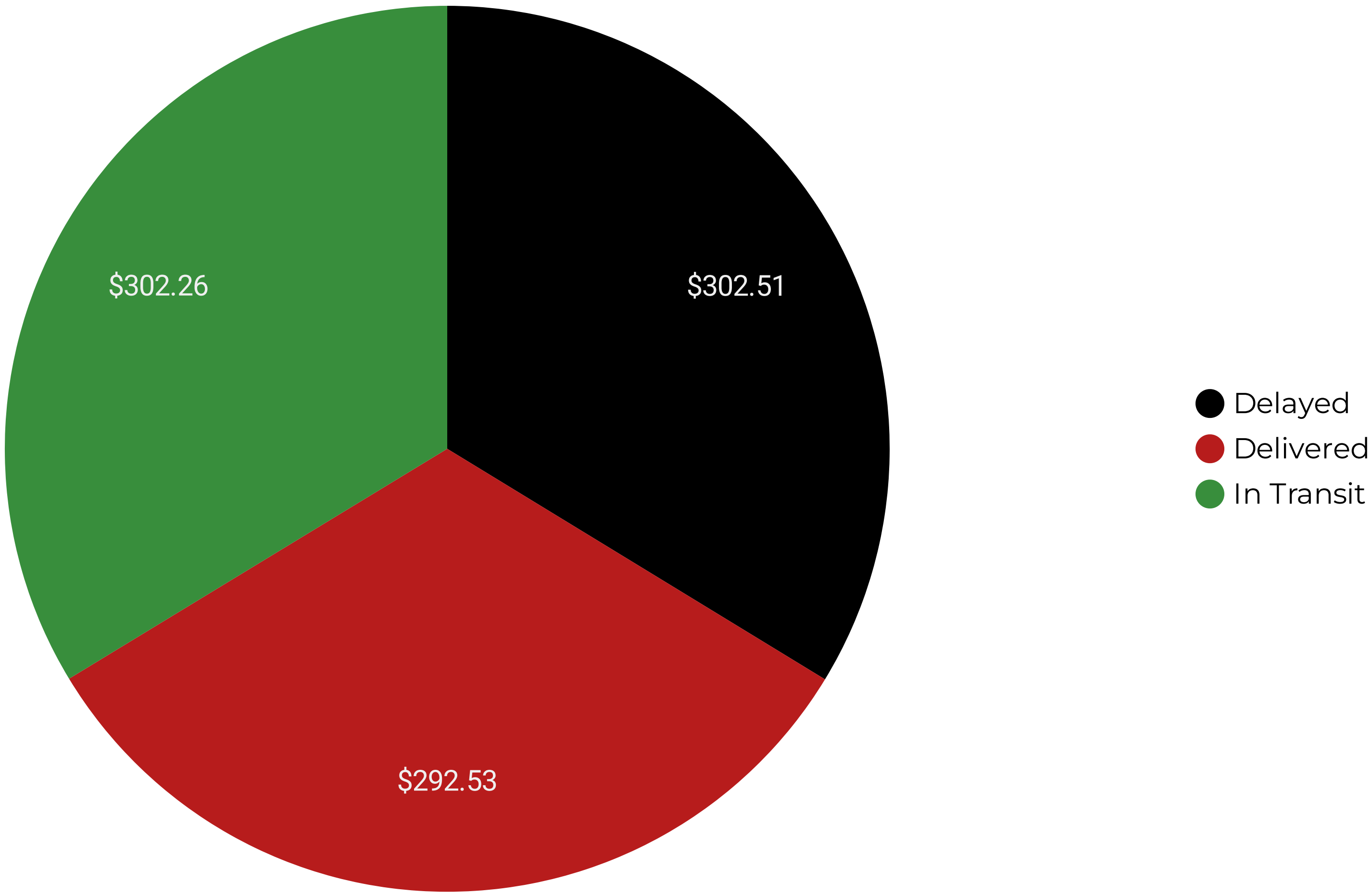
From the environment analysis we can see that with delay the average humidity is 65.04 & average temperature of 23.78. But, with no delay we can see the average humidity is 65.05 & average temperature of 24.04. However, the weather looks normal enough for the trucks. Meaning, these weather conditions do not have an impact towards the data.

Customer & Sales



So what is the main problem? Well from everything we saw the best way to fix this company issue is to reroute shipments before they enter heavy-traffic zones because this is the single highest-leverage lever available in the data. Next they need to start flagging the shipments and notify the customers since the average wait time is over 35 minutes during every type of traffic. By doing all of this it will also fix the delay in sales, and vehicles that are in transit already.